0. what's your hardest technical challenge.

- Represented the state-transition rules of a distrete dynamical system in first-order logic in a way that could used to represent all possible initial states and evolutions of the system.
- Extended a computer algebra system to manipulate symbolic logic expressions.
- used the extended CAS to translate the FOL expressions to Boolean logic expressions such that they could instantiated and inputted into a SAT Solver.
- Used the solver to find initial configurations that evolve into a specified final state after any number of generations.
- In practice this meant I could take any state of Conway's game of life and search for an initial state that evolved into that state after an arbitrary number of generations.
  - show video on laptop.
- There were two hard things that I solved:
  - Converting the systems rules into first order logic and figuring out how to manipulate them algebraically.
  - extending a CAS to manipulate symbolic logic expressions. The only reason I was able to do this was that I'd spent a few years becoming an expert at the Wolfram Language.

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I extended a computer algebra system to handle symbolic logic so I could encode the
state-transition rules of Conway's Game of Life in first-order logic. Then I
translated those expressions into Boolean logic suitable for a SAT solver. The
solver let me find initial configurations that evolve into a specified final state—
like a board spelling my professor's name. The hardest parts were representing the
rules algebraically and modifying the CAS to manipulate first-order logic.
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0b. What's your biggest software engineering challenge.

- the hardest problems i solved where in maths. But if you want a software engineering problem I solved recently I have one.
- So at work I was given 1 day to solve the following:
  - Write a DB-less endpoint that discovers a wallet's ERC-4626 vault positions across the entire chain in ~10s.
  - Used eth_getLogs with topic0 = keccak256("Deposit(address,address,uint256,uint256)") and topic2 = user to find deposits without preknown vault addresses.
  - Sharded the chain into block ranges and ran bounded-concurrency scans with retries

1. What's a Hash function?

A function that converts an input of any size into an output of a fixed lenght called a hash.

- This Hash is a digital finger print. There is usually 1 unique for each input of a good hash function.

2. What’s a Cryptographic Hash function

A Hash function with 2 properties.

Pre-image resistance: given the hash it’s infeasible to find the input.

Collision Resistance: Two inputs are never mapped to the same output.

3. What’s a Hash Table

A data structure consisting of a set of key value pairs that offers very fast lookup. A hash table has two common operations: insert, lookup.

To lookup a value you can hash the key: The hash gives you the index of the corresponding value in a bucket array.

To insert a key-value pair. You also hash the key. This gives you an index or memory address for a particular bucket in the bucket array in which to put the key value pair.

Hash[k] = memory_address_of_value.

It’s created by hashing all the keys. To get a set of indexes in a bucket array.

- worst case: O(n). Average case: O(1)

3. How do I make a Hash table collision resistant.

linear probing. When a collision occurs you attempt to insert the key value pair in a neighbouring bucket.

- chaining -> putting a list of key value pairs in a bucket.

4. what is symmetric encryption

- Symmetric encryption uses a single shared key for both encryption and decryption.

5. what is assymmetric encryption

- Asymmetric encryption uses two related keys: a public key (shared) and a private key (kept secret).
- you don’t need to share the secret to decrypt the message.

6. what is the difference between symmetric and assymmetric encryption?

7. What is my motivation to join Sony?

- blockchain is still a very nascent technology.
- currently it’s the purview of small scrappy startups
- I think the space needs corporate participation in order to have a big impact on humanities future.
  - The issue with working at a startup is that we didn’t have the scale to execute ambitious plans. No one trusted us enough to build on top of us because we were a small startup.
- I love blockchain technology. why?

8. What do you love blockchain technology?

- It enables cooperation between humans on a planetary scale.
- It’s a planetary technology. If we want to expand extropically into the cosmos and become a Kardashev type 2 civilization. It’s essential that we have systems that align our incentives and allow us to cooperate.
- like bees in a beehive.

- It sits at the intersection of **incentives, economics, and computation**.
- It models **distributed systems**, which I’ve studied and admired—from neural networks to human societies.
- It enables **large-scale cooperation** without central authority.
- It serves as a **technological substrate for collective intelligence**, much like a hive mind.
- It represents an early step toward **Kardashev Level 1 civilization**—a humanity capable of coordinating effectively across the entire planet.

9. What is a digital signature?

It proves too things:

- message came from alice
- message was not altered. (hash shows this)

10. how does it work?

- Alice hashes her message
- encrypts it with her private key
- sends this signature + the original message to bob
- bob can then hash the original message
- use alice’s pubic key to decrypt the signature into a hash
- compare the two hashes

11. What’s a layer 2 scaling solution?

- mainnet can only handle low double digit transactions per second. How do we scale to a planetary financial system.

12. What’s an optimistic roll up?

13. What’s the Ethereum memory model?

- consists of: storage, memory, calldata, stack
- storage is permanent, expensive, used for storing, balances, addresses
- memory stores temporary data while a contract executes.
- Call-data: the function + arguments. that are sent to the smart contract by a user to execute a function of that smart contract
- stack: temporary values during computation: like the CPU stack in classical computing: LIFO

14: what’s impermanent loss?

loss relative to just holding the tokens outside the pool. price moves away from the tick range. but price can move back. hence impermanent.

15: Whats an optimistic roll up?

- we assume tx’s are valid optimistically.
- challenge window to prove fraud. anyone can submit a fraud proof

16: What’s a ZK roll up?

- It’s an off-chain program that processes transactions.
- It posts a ZK proof that it processed the transaction correctly to mainnet. All mainnet has to do is verify the proof.

17: What’s the difference between a smart contract and a regular program?

- state is public and immutable.
- smart contract is a subset of the space of all programs
  - deterministic
  - runs on the EVM
  - costs gas to run
  - triggered by a user or other smart contract executing transactions

Ethereum dev guide Questions

18. What is a blockchain, in the context of Ethereum?

- So, a blockchain, in the context of Ethereum, is a public distributed ledger that’s copied onto a bunch of nodes in the Ethereum network.
- And this ledger consists of blocks, which are data structures containing timestamped transactions that all occur in a timestamped sequence.
- And each block is linked, it contains the hash of the header of the previous block, such that it’s impossible to modify past blocks without changing all the blocks after it, such that Ethereum is an append-only public database, append-only and immutable.

19. How does Ethereum’s consensus mechanism currently work?

- proof of stake. special nodes, validators stake eth, they get slashed if they do bad stuff. a node is chosen at random to propose a block. if 2 thirds of the network agree then the block is appended.

20. What is the role of Ether (ETH) in the Ethereum

network?

- GAS: pay for compute, stop spam as a side effect.
- STAKE: used to secure the Proof of stake consensus mechanism. Validators risk slashing for bad behaviour
- CURRENCY: medium of exchange, store of value in DeFi.

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21. Define a “smart contract” on Ethereum.

- compiled to bytecode and stored on-chain
- a program which runs on the Ethereum network on the Ethereum virtual machine. And this program has a shared state which is public, it’s immutable, it is triggered by executing transactions, it’s also deterministic.
- provides functions which can be called by off chain programs by sending call data to their address on a node.

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22. What is the Ethereum Virtual Machine (EVM)?

- executes EVM bytecode deterministically
- is a run time environment or substrate on which smart contracts can be run.
- costs gas
- necessary resources that allow smart contracts to run, like storage, memory, the stack

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23: What are transactions in Ethereum — what do they do and how are they executed?

- transaction is a signed message sent from an EOA that requests a state change. e.g. sending ETH, calling a contract
- executed sequentially by the EVM
- Their results are included in blocks; once a block is finalized, the transaction’s effects become part of the global state.

24: What is an account in Ethereum and how is it different from a smart contract?

- An account in Ethereum is an entity with an address, balance, nonce, and storage root.
- Two types:

    - EOA’s, controlled by a private key, can sign and send transactions.
    - Smart Contract account. Controlled by code stored on chain. Can execute transactions when called.

25: What are blocks in Ethereum and how do they relate to transactions and nodes?

- A block is a data structure containing:
    - [tx1, tx2, tx3...]
    - Mon 27 Oct 2025 15:18:39 GMT+1
    - reference to previous block. hash of header.
- Validators propose and attest to new blocks under Proof of Stake.
- Nodes store, verify, and propagate blocks, maintaining consensus on the canonical chain.

26: How do nodes function in the Ethereum ecosystem?

- Nodes run Ethereum client software that implements the protocol rules.
- They store and sync the blockchain’s full or partial state.
- They verify blocks and transactions to ensure validity.
- They propagate new transactions and blocks across the peer-to-peer network.
- Some nodes act as validators, proposing and attesting to blocks; others act as non-validating full nodes or light clients that serve data and maintain trustless access to the chain.

27: Why does Ethereum charge fees (gas) for computation and state changes?

- to pay nodes to maintain compute resources
- to stop spam
- to align incentives, and make network more efficient. block space is a finite resource and must be allocated via a market.
- the network remains secure, efficient, and resistant to abuse.

28: Suppose you want an auction with private bids hidden until the end. How do you prevent others from

seeing or frontrunning those bids on-chain?

- unfounded question
- You can’t hide the existence of events on chain. Instead do this off-chain.
- Instead of completely hiding the bids one could encrypt them and store them on chain as encrypted bids. But they could still be front run.

29: What does MEV stand for?

- Maximal Extractable Value.

30: Explain what Aave is?

- A DeFi lending protocol, users can deposit, or borrow against collateral.
- interest rates are algorithmically determined by supply and demand in each liquidity pool
- atokens

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31. how does Aave work?

- If collateral value falls below the health factor (a threshold), liquidators repay the debt and claim the collateral via smart contracts.

32: Explain what Uniswap is?

- Uniswap is a DEX

33: how does uniswap work?

- fees are collected by liquidity providers. determined by the ppool contract.

34: How is the base fee for gas on Ethereum calculated

- as a function of the fee in the previous block.

35: How to define something.

- What it does. sign transactions, roar like a duck.
- What it is. Datastructure or entity.

ssimplest zero knoweldge proof

prover knows a secret x=5;

verifier tries to verify that the prover knows x

In[ ]:=
```
(*simple zero-knowledge proof setup*)
x = 5;        (*secret*)
g = 3;        (*generator*)
p = 23;        (*prime modulus*)

y = PowerMod[g, x, p];  (*public key*)

r = 7;           (*random nonce*)
a = PowerMod[g, r, p];  (*commitment*)

c = RandomInteger[{0, 1}];  (*verifier challenge*)

z = Mod[r + c * x, p - 1];   (*prover’s response*)

(*verifier check*)
lhs = PowerMod[g, z, p];
rhs = Mod[a * PowerMod[y, c, p], p];

<|"Challenge" → c, "Verified" → (lhs == rhs) |>
```

- viktor the verifier tells peggy to exit from A or B. if he asks her to exit through B, and she has gone down A. and she doesn’t know the password for the door then she will have to go back down A. And bob will see.

make demo notebook

minimal demo of applying de morgans to a logic expression

In[ ]:=
```
deMorganExpand[expr_] := expr //. {not[and[a_, b_]] :→ or[not[a], not[b]],
        not[or[a_, b_]] :→ and[not[a], not[b]]}
```

In[ ]:=
```
DeMorganExpand[not[and[a, b]]]
```

Out[ ]=
```
or[not[a], not[b]]
```

I supplied common propositional equivalences as extra term re-write rules in mathematica. I added them to the system. Extending it.

What is attestation?

An attestation in Proof of Stake is a validator’s signed vote confirming that a specific block and chain view are valid. It’s used to reach consensus and finalize blocks.

what’s a tx

a signed message that requests a change in state of the network.